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ELECTRO-MYOGRAPHY

Every movement begins as an electrical signal. This demo lets you control a game using the signals measured from your muscles.



Electromyography (EMG) is a technique used to measure the electrical activity produced by skeletal muscles.



When your brain sends a signal to contract a muscle, motor neurons activate groups of muscle fibres. This activation generates tiny electrical signals known as muscle action potentials.



Surface EMG records these signals using electrodes placed on the skin above the muscle.



The strength of the EMG signal generally increases as more motor units are recruited and muscles produce greater force.

EMG is widely used in biomechanics, rehabilitation, sports science, prosthetics, and human-computer interaction: It provides insight into how the nervous system controls movement.

How does this demo work?

Electrodes placed on the skin detect your muscle activation in real time. The EMG signal is then converted into a control input for the game: The stronger you contract, the higher the bird will fly!



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